

BOOLEAN FUNCTIONS IN CRYPTOGRAPHY

ABSTRACT. These are preliminary lecture notes for a course on Cryptographic Boolean Functions.

1. INTRODUCTION

Many block and stream ciphers employ Boolean functions $\mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$ as primitives. For these symmetric ciphers one does not usually have exact descriptions of cryptographic security. However, the existing attacks on them describe various and usually contradictory definitions of cryptographic security. These include

- **Differential attacks** and **differential uniformity**,
- **Linear attacks** and **non-linearity**,
- **Algebraic attacks** and **degree/algebraic immunity**,
- **Correlation attacks** and **correlation immunity**.

We assume that the basics of two of these attacks (differential attacks and linear attacks) are known to the reader and concentrate on the corresponding mathematical definitions for Boolean functions, i.e., differential uniformity and non-linearity. We investigate optimal functions that arise from these definitions (**perfect nonlinear** and **bent** functions). Our goals include showing existence of such functions when $k \leq n/2$ and proving equivalence of these two definitions.

We are also interested in related combinatorial objects, such as **Latin squares**, **Hadamard matrices**, **orthogonal arrays**, **difference sets**, **block designs** and **projective planes**. We are going to show that how the cryptographic definitions lead to these objects and then we will conversely show how other cryptographic applications arise from these combinatorial objects, such as Latin squares and secret-sharing-schemes. We will also briefly mention the relationship of Boolean functions with **error-correcting codes**.

We are mostly interested in the Boolean case (i.e., functions are over **binary** vector spaces) but the p -ary versions (i.e., functions $\mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$ where p is odd) will also be explored in some detail.

2. FINITE FIELDS

A field is a set F together with two operations $(+, \cdot)$ that satisfy the following axioms:

- (i) associativity of both $+$ and $\cdot$,
- (ii) commutativity of both $+$ and $\cdot$,
- (iii) existence of additive identity 0 and multiplicative identity 1 of $+$ and $\cdot$ respectively,
- (iv) existence of additive inverses $-a$ and multiplicative inverses $1/b$ with respect to $+$ and $\cdot$ of $a, b \in F$ with $b \neq 0$,
- (v) distributivity of $\cdot$ over $+$.

Equivalently, $(F, +)$ and $(F \setminus \{0\}, \cdot)$ are commutative groups together with the axiom (v). This amounts to saying that F is a non-zero (commutative) ring with multiplicative inverses. We define $F^\times = F \setminus \{0\}$.

We will shortly see that the additive group must be elementary abelian, which is equivalent to saying that it is a finite vector space. We are interested in finite fields since the cryptographically interesting functions are defined over finite vector spaces.

2.1. The prime subfield. Let F be a finite field. The multiplicative identity $1_F \in F^\times$, generates a finite subfield E of F since $n1_F = 1_F + 1_F + \cdots + 1_F = 0$ for an integer n (additive order is finite).

Remark 2.1. (i) Recall that subrings of fields are integral domains since a solution of $ab = 0$ in the subring is a solution in the field and $ab = 0$ in a field implies $a^{-1}ab = a^{-1}0$ which in turn implies $b = 0$ or $a = 0$.

(ii) Recall that every finite integral domain is a field since $ab_1 = ab_2$ implies $a(b_1 - b_2) = 0$ hence multiplication by a is injective and therefore surjective since F is finite. Thus $ac = 1$ for some $c \in F$.

Therefore the homomorphism from $\mathbb{Z} \rightarrow F$ defined by

$$(1 + 1 + \cdots + 1) = n1 \mapsto n1_F = (1_F + 1_F + \cdots + 1_F)$$

must have as kernel a maximal ideal in $\mathbb{Z}$. The ideals of $\mathbb{Z}$ are $m\mathbb{Z}$ for $m \in \mathbb{Z}$ and they are maximal if $m = p$ is a prime. Thus F has a subfield isomorphic to $\mathbb{Z}/(p)$ which we call $\mathbb{F}_p$. Since a field is a vector space over any of its subfields and F is finite, it is then a (finite dimensional) vector space over $\mathbb{F}_p$. Thus

$$F = \{c_1\alpha_1 + \cdots + c_n\alpha_n : c_1, \dots, c_n \in \mathbb{F}_p\},$$

where $\alpha_1, \dots, \alpha_n$ is a basis of F over $\mathbb{F}_p$. By counting, we prove

Theorem 2.2. *A finite field has order p^n with $n \in \mathbb{N}$.*

Remark 2.3. Check the vector space axioms to see that the field axioms indeed imply the vector space axioms.

The rings $\mathbb{Z}$ and $\mathbb{F}_p[x]$ are analogous in a very important way: They are both Euclidean domains, i.e., they are integral domains that admits a Euclidean norm function that leads to a Euclidean division algorithm. The norm function is the absolute value for $\mathbb{Z}$ and the degree in $\mathbb{F}_p[x]$, as you might recall. Also recall that, primes in $\mathbb{F}_p[x]$ are those irreducible polynomials, that are monic (i.e., leading coefficient is 1) polynomials that do not admit non-trivial divisors (i.e., only units and itself). Thus, the maximal ideals of $\mathbb{F}_p[x]$ are (f) for irreducible f . This gives us a method to generate finite fields with prime power order, that is, $F \cong \mathbb{F}_p[x]/(f(x))$ for irreducible f . Note that the Euclidean algorithm implies for any $0 \neq a \in \mathbb{F}_p[x]$ with degree $\deg a < \deg f$, the existence of c, d such that.

$$ac + fd = 1,$$

which means $ac \equiv 1 \pmod{f}$ which shows the existence of inverses in the quotient ring, essentially the same as the proof for integer primes. Such extensions are called algebraic extensions. Let α be a root of the irreducible polynomial

$$f = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in \mathbb{F}_p[x].$$

Thus, α satisfies an algebraic equation of degree n and is said to be algebraic (f is called the minimal polynomial of α). Then

$$\mathbb{F}_p(\alpha) = \{c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1} : c_i \in \mathbb{F}_p\}$$

is a field (with natural multiplication and the vector space addition) isomorphic to $\mathbb{F}_p[x]/(f(x))$. Note that multiplication is reduced using

$$\alpha^n = -a_{n-1}\alpha^{n-1} - \cdots - a_1\alpha - a_0.$$

We call this process adjoining a root of a polynomial to a field.

The field F has order p^n if $\deg f = n$ for irreducible f . The question is whether there exists an irreducible for every p and n .

We cite the following results from field theory without proof. They are mostly straightforward and can be found in your favourite algebra book.

Theorem 2.4. *(i) Every finite extension of a field is algebraic.*

(ii) Let $F \subseteq K \subseteq L$ be field extensions. Then the degrees satisfy $[L : F] = [L : K][K : F]$.

(iii) For every polynomial $g \in F[x]$ a splitting field K exists (i.e., g splits in $K[x]$ into linear factors) and is unique up to isomorphism.

We are now in a position to prove existence and uniqueness of finite fields of every possible order.

Theorem 2.5. *Let p be a prime and $n \in \mathbb{N}$. The splitting field of $x^{p^n} - x \in \mathbb{F}_p[x]$ has exactly p^n elements.*

Proof. Let F be the splitting field in question and

$$E = \{a \in F \mid a^q - a = 0\}.$$

It is easy to show that E itself is a field and therefore $E = F$. We just show $a \in E$ and $b \in E$ implies (closure) $ab = a^qb^q = (ab)^q$ and (inverses) $a/b = a^q/b^q = (a/b)^q$. Also $a \pm b = a^q \pm b^q = (a \pm b)^q$ using the well known binomial identity modulo p . The derivative of $f = x^q - x$ is $f' = p^n x^{p^n-1} - 1 = -1$. This means $(f, f') = 1$ which implies that there are no double roots completing the proof. $\square$

Note that by Theorem 2.4 (iii), the finite field of order p^n is unique, which we write $\mathbb{F}_{p^n}$.

Theorem 2.6. *The subfields of $\mathbb{F}_{p^n}$ are of the form $\mathbb{F}_{p^d}$ where $d \mid n$. Each of these subfields $\mathbb{F}_{p^d}$ is unique.*

Proof. If a subfield E (of order p^d) of F (of order p^n) exists, then F is an E -vector space thus $d \mid n$. Conversely, if $d \mid n$ then $p^d - 1 \mid p^n - 1$ implying $x^{p^d-1} - 1 \mid x^{p^n-1} - 1$. All elements $E = \mathbb{F}_{p^d}$ should satisfy $x^{p^d} - x = 0$, thus it must be unique in F . $\square$

Now we have a complete picture of the factorization of $x^{q^n} - x \in \mathbb{F}_q[x]$:

Theorem 2.7.

$$x^{q^n} - x = \prod_{d \mid n} \prod_{f \in I_q(d)} f,$$

where $I_q(d)$ denotes the set of monic irreducibles in $\mathbb{F}_q[x]$ of degree d .

Proof. By induction on the prime factorization of n using Theorem 2.4 (i) and Theorems 2.5 and 2.6. That is to say, the elements of $\mathbb{F}_{p^n}$ are generated by irreducible polynomials of degree d for all $d \mid n$. $\square$

2.2. The multiplicative group. Recall the structure theorem of finite abelian groups.

Theorem 2.8. *Let G be a finite abelian group. Then*

$$G \cong \mathbb{Z}_{p_1^{k_1}} \times \mathbb{Z}_{p_2^{k_2}} \times \cdots \times \mathbb{Z}_{p_r^{k_r}}$$

where p_i are not necessarily distinct. And,

(i) G is cyclic if $p_i \neq p_j$ for $i \neq j$,

(ii) G is elementary abelian if $p_i = p_j$ and $k_i = 1$ for all i and j .

We will now see the additive and multiplicative groups of F are the two extremes.

Theorem 2.9. *The multiplicative group $(F^\times, \cdot)$ of a finite field F is cyclic.*

Proof. Let F be a finite field with q elements and $F^\times \cong A \times B$, i.e., $F^\times$ can be written as a direct product of two groups, with $|A| = a$ and $|B| = b$. An element (g, h) of $A \times B$ has order dividing $\text{lcm}(a, b)$ since

$$(g^{ab/\text{gcd}(a,b)}, h^{ab/\text{gcd}(a,b)}) = (g^{a(b/\text{gcd}(a,b))}, h^{b(a/\text{gcd}(a,b))}) = (1, 1).$$

For all $x \in F^\times$ we have $x^{\text{lcm}(a,b)} = 1$, or equivalently for all $x \in F$ we have $x^{\text{lcm}(a,b)+1} - x = 0$. A polynomial f in $F[x]$ has at most $\deg f$ solutions, thus $x^q - x$ has at most $\text{lcm}(a, b) + 1$ solutions. Thus $\text{lcm}(a, b) = ab = q - 1$, or $\text{gcd}(a, b) = 1$. By the structure theorem of finite abelian groups $F^\times$ is cyclic. $\square$

2.3. Automorphism of finite fields. Automorphisms of the multiplicative group $(\mathbb{F}_q^\times, \cdot)$ (which is cyclic $\cong \mathbb{Z}/(q-1)$) and the additive group $(\mathbb{F}_q, +)$ (which is elementary abelian $\cong \mathbb{Z}/(p) \times \cdots \times \mathbb{Z}/(p)$) is our next topic.

Note that an automorphism ϕ of $(\mathbb{F}_q^\times, \cdot) = \langle g \rangle$ satisfies $\phi(g^j) = \phi(g)^j$ for all j thus $\phi : x \mapsto x^d$ for some $(d, q-1) = 1$. Hence $\text{Aut}(\mathbb{F}_q^\times, \cdot) \cong (\mathbb{Z}/(q-1))^\times$. Since an elementary abelian p -group of order $q = p^n$ is an $\mathbb{F}_p$ -vector space of dimension n , we know from linear algebra that $\text{Aut}(\mathbb{F}_q, +)$ comprises vector space automorphisms which are of type $x^T \mapsto Ax^T$ where A is a nonsingular $n \times n$ matrix over $\mathbb{F}_p$ and denoted $\text{GL}(n, p)$.

All endomorphisms of the multiplicative group are $x \mapsto x^d$ for any $1 \leq d \leq q-1$ (i.e., $\cong \mathbb{Z}/(q-1)$); and the endomorphism ring of the additive group is $\mathbf{M}_n(\mathbb{F}_p)$ (i.e., $\mathbb{F}_p$ -algebra of $n \times n$ not necessarily non-singular matrices over $\mathbb{F}_p$) respectively.

Now, let us consider the field automorphisms $\text{Aut}(\mathbb{F}_q, +, \cdot)$. It is immediate from the identity

$$(x + y)^{p^i} = x^{p^i} + y^{p^i}$$

in $\mathbb{F}_q$, that $\sigma : x \mapsto x^p$ generates a cyclic group of order n of $\mathbb{F}_q$ -automorphisms. Now $\text{Aut}(\mathbb{F}_q, +, \cdot) = \langle \sigma \rangle$ follows from the following fact from Galois theory.

Theorem 2.10. *Let E be the splitting field over F of the polynomial $f \in F[x]$. Then $|\text{Aut}(E/F)| \leq [E : F]$ with equality if f is separable over F .*

In the theorem $\text{Aut}(E/F)$ stands for the group of automorphisms of E that fixes F (pointwise), i.e.,

$$\text{Aut}(E/F) = \{\sigma \in \text{Aut}(E) : \sigma(a) = a, \forall a \in F\}.$$

If $F = \mathbb{F}_p$ then obviously $\text{Aut}(E) = \text{Aut}(E/F)$.

3. FUNCTIONS OVER FINITE FIELDS

3.1. Vectorial functions. Recall that our main aim is to understand functions over finite fields. For the multiplicative automorphisms we used exponential maps whereas for additive ones matrices were employed. We will show that we can unify both by making use of polynomial functions. In finite fields every function can be interpreted as an evaluation of a polynomial as we will now see.

Let $\mathcal{F}(\mathbb{F}_q^n, \mathbb{F}_q)$ denote the $\mathbb{F}_q$ -algebra (i.e., $\mathbb{F}_q$ -vector space with respect to function addition endowed with an $\mathbb{F}_q$ -bilinear function multiplication) of functions $\mathbb{F}_q^n \rightarrow \mathbb{F}_q$. We will show that

$$R_{q,n} = \mathbb{F}_q[x_1, \dots, x_n] / (x_1^q - x_1, \dots, x_n^q - x_n) \cong \mathcal{F}(\mathbb{F}_q^n, \mathbb{F}_q)$$

as $\mathbb{F}_q$ -algebras. Let us start with the case $n = 1$. First, observe that for $x \in \mathbb{F}_q$ we have $x^q = x$. What this means is that every function f defined on $\mathbb{F}_q$ elements that takes on $\mathbb{F}_q$ values can be written (uniquely) as polynomials of the form

$$\tilde{f}(x) = \sum_{i=0}^{q-1} a_i x^i, \quad a_i \in \mathbb{F}_q.$$

preserving addition and multiplication of functions. Note that the function f is the evaluation function $\text{ev}(\tilde{f})$ of the polynomial $\tilde{f}$, and ev is obviously an algebra homomorphism. Given that the cardinalities of the two sets are both q^q . It is enough to show surjectivity or injectivity. We can show both easily.

- (i) Kernel of ev is trivial which can be proved by the observation that if $f - g$ is the zero function, then $\tilde{f} - \tilde{g}$ must be the zero polynomial, since otherwise we would have a nonzero polynomial with degree $< q$ with q roots.
- (ii) For the injectivity, we will introduce a fundamental notion known as the Lagrange interpolation. That is to say, every function can be written as a polynomial with degree $< q$. Let

$$\delta_a(x) = \begin{cases} 1 & \text{if } x = a, \\ 0 & \text{if } x \neq a. \end{cases}$$

It is easy to see that evaluation of

$$\tilde{\delta}_a(x) = 1 - (x - a)^{q-1} = \prod_{b \in \mathbb{F}_q \setminus \{a\}} \frac{x - b}{a - b}$$

is the function we are looking for, and

$$\tilde{f}(x) = \sum_{a \in \mathbb{F}_q} \tilde{\delta}_a(x) f(a).$$

3.2. Linearized polynomials. Recall that the multiplicative endomorphisms of $\mathbb{F}_q^\times$ are of the form $x \mapsto x^d$ where $1 \leq d \leq q-1$. Thus, they can be viewed as evaluation maps arising from monomials in $R_{q,1}$ (which additionally maps $0 \mapsto 0$). For the additive endomorphisms $x^T \mapsto Ax^T$, define

$$\mathcal{L}_{p,n} = \{a_0x + a_1x^p + a_2x^{p^2} + \cdots + a_{n-1}x^{p^{n-1}} : a_i \in \mathbb{F}_{p^n}\}.$$

Observe that all polynomials $\tilde{L}$ in $\mathcal{L}_{p,n}$ define $\mathbb{F}_p$ -linear maps $L : \mathbb{F}_q \rightarrow \mathbb{F}_q$, that is to say,

- (i) $L(x) + L(y) = L(x + y)$ for all $x, y \in \mathbb{F}_q$, and
- (ii) $L(cx) = cL(x)$ for all $c \in \mathbb{F}_p$ and $x \in \mathbb{F}_q$,

recalling that $\mathbb{F}_q$ is an n -dimensional $\mathbb{F}_p$ -vector space. This is an easy deduction from the Frobenius identity for all $i \in \mathbb{N}$:

$$(x + y)^{p^i} = x^{p^i} + y^{p^i}.$$

Note that there are $q^n = p^{n^2}$ linearized polynomials in $\mathcal{L}_{p,n}$ that describe pairwise different linear endomorphisms of the vector space $(\mathbb{F}_q, +)$ (recalling that the difference of two different polynomials describe a nonzero map). Thus, every linear transformation of $\mathbf{M}_n(\mathbb{F}_p)$ can be realized in $\mathcal{L}_{p,n}$.

One can prove the following assertion.

Theorem 3.1. *The algebra of $\mathbb{F}_p$ -linearized polynomials together with addition and composition of polynomials and the matrix algebra $\mathbf{M}_n(\mathbb{F}_p)$ together with matrix addition and multiplication are isomorphic $\mathbb{F}_p$ -algebras, i.e.,*

$$(\mathcal{L}_{p,n}, +, \circ) \cong (\mathbf{M}_n(\mathbb{F}_p), +, \cdot).$$

Proof. (Sketch) Observe that $L_1, L_2 \in \mathcal{L}_{p,n}$ implies $L_1 + L_2, L_1 \circ L_2 \in \mathcal{L}_{p,n}$ reducing modulo $x^q - x$ and compatibility with $\mathbb{F}_p$ and distributivity is trivial. Pick a basis $\{\beta_0, \dots, \beta_{n-1}\} \subset \mathbb{F}_q$ for $\mathbb{F}_q$ over $\mathbb{F}_p$ and identify vectors as

$$x = \sum_{i=0}^{n-1} x_i \beta_i.$$

Plugging in the basis describes a homomorphism between the two algebras (proof of which requires straightforward but lengthy computations). The underlying sets have the same cardinality as shown before the theorem. $\square$

Example 3.2. Let an irreducible polynomial $f \in \mathbb{F}_q[x]$ be defined as

$$f(x) = x^k - a_{k-1}x^{k-1} - a_{k-2}x^{k-2} - \cdots - a_0.$$

Let f be the minimal polynomial of α over $\mathbb{F}_q[x]$ which is the unique monic irreducible polynomial of degree k satisfying $f(\alpha) = 0$. Now, $B = \{1, \alpha, \alpha^2, \dots, \alpha^{k-1}\}$ is a basis

of $\mathbb{F}_{q^k}$ over $\mathbb{F}_q$. Next, consider the following $k \times k$ matrix A with $\mathbb{F}_q$ -coefficients, known as the **companion matrix** of f , defined as

$$A = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & a_0 \\ 1 & 0 & 0 & \cdots & 0 & a_1 \\ 0 & 1 & 0 & \cdots & 0 & a_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & a_{k-1} \end{pmatrix}.$$

View any $x \in \mathbb{F}_{q^k}$ as the vector $(x_0, \dots, x_{k-1}) \in \mathbb{F}_q^k$ by writing x in the basis B , i.e., $x = \sum_{i=0}^{k-1} x_i \alpha^i$. It is easy to show that the linear map induced by the matrix A , i.e., the map

$$A : (x_0, \dots, x_{k-1})^t \mapsto A(x_0, \dots, x_{k-1})^t,$$

represents the $\mathbb{F}_{q^k}$ -linear (and hence $\mathbb{F}_q$ -linear) map defined by

$$m_\alpha : x \in \mathbb{F}_{q^k} \mapsto \alpha x \in \mathbb{F}_{q^k},$$

relative to the basis B .

Exercise 3.3. (i) Perform the computations by keeping in mind that $\mathbb{F}_{q^k} \cong \mathbb{F}_q[x]/(f)$.

(ii) Describe m_β for any $\beta \in \mathbb{F}_{q^k}$. That is to say describe all matrices that represent multiplication by β in the basis B .

(iii) Show that the set

$$\{m_\beta : \beta \in \mathbb{F}_{q^k}\}$$

is isomorphic to the field $\mathbb{F}_q[x]/(f)$ or equivalently $\mathbb{F}_q(\alpha)$.

(iv) Make sure that you remember the notions **characteristic polynomial**, **trace**, **determinant** of square matrices. How are these notions of m_α are related to some properties of α ?

3.3. Vectorial functions as polynomials. Now let us give the full theorem (i.e., general n) for vectorial functions.

Theorem 3.4. *The $\mathbb{F}_q$ -algebra of vectorial functions $\mathcal{F}(\mathbb{F}_q^n, \mathbb{F}_q)$ (together with function addition and multiplication) is isomorphic to $R_{q,n}$ (together with polynomial addition and multiplication), i.e.,*

$$R_{q,n} = \mathbb{F}_q[x_1, \dots, x_n]/(x_1^q - x_1, \dots, x_n^q - x_n) \cong \mathcal{F}(\mathbb{F}_q^n, \mathbb{F}_q)$$

Proof. (Sketch) The assertion that the usual evaluation map

$$\text{ev} : R_{q,n} \rightarrow \mathcal{F}(\mathbb{F}_q^n, \mathbb{F}_q)$$

is a homomorphism is straightforward. We only need to give a bijection, which will be given by a generalized Lagrange interpolation, i.e.:

$$\delta_{\underline{a}}(\underline{x}) = \begin{cases} 1 & \text{if } \underline{x} = \underline{a}, \\ 0 & \text{if } \underline{x} \neq \underline{a}. \end{cases}$$

Similarly, as before

$$\tilde{\delta}_{\underline{a}}(\underline{x}) = \prod_{i=1}^n (1 - (x_i - a_i)^{q-1}) = \prod_{i=1}^n \prod_{b_i \in \mathbb{F}_q \setminus \{a_i\}} \frac{x_i - b_i}{a_i - b_i}$$

is the function we are looking for, and

$$\tilde{f}(\underline{x}) = \sum_{\underline{a} \in \mathbb{F}_q^n} \tilde{\delta}_{\underline{a}}(\underline{x}) f(\underline{a}).$$

□

3.4. The notion of the algebraic degree. A function $f \in \mathcal{F}(\mathbb{F}_q^n, \mathbb{F}_q)$ is a function that maps n $\mathbb{F}_q$ -symbols to one $\mathbb{F}_q$ -symbol. Let $q = p^m$. So, fixing a basis of $\mathbb{F}_q$ over $\mathbb{F}_p$, the same function f can be viewed as a function that maps nm $\mathbb{F}_p$ -symbols to m $\mathbb{F}_p$ -symbols, i.e., $(\mathcal{F}(\mathbb{F}_p^{nm}, \mathbb{F}_p))^m$. Note that this is an $\mathbb{F}_p$ -vector space isomorphism (writing $\mathbb{F}_q$ in an $\mathbb{F}_p$ -basis). For the purpose of the degree discussion let $f \in (\mathcal{F}(\mathbb{F}_p^n, \mathbb{F}_p))^k$ or simply $f : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$ (not necessarily $k|n$). So we can view

$$f(\underline{x}) = (f_1(\underline{x}), f_2(\underline{x}), \dots, f_k(\underline{x})),$$

with each $f_i : \mathbb{F}_p^n \rightarrow \mathbb{F}_p$. Thus we define the **algebraic degree** of f as the maximum of the degrees of the **coordinate functions** f_i . We say that a linear combination $\langle c, f \rangle = \sum_{j=1}^n c_j f_j$ is a **component function** of f .

Let us write down the two definitions of degree formally.

Definition 3.5. (Polynomial degree) Let $f \in \mathcal{F}(\mathbb{F}_q^n, \mathbb{F}_q)$ (written in $R_{q,n}$) be

$$\tilde{f}(x_1, \dots, x_n) = \sum_{i_1, \dots, i_n=0}^{q-1} a_{i_1, \dots, i_n} \prod_{j=1}^n x_j^{i_j}.$$

Then

$$\deg_p f = \max\{i_1 + \dots + i_n : a_{i_1, \dots, i_n} \neq 0\}$$

Definition 3.6. (Algebraic degree) Let $\underline{f} = (f_1, \dots, f_k) \in (\mathcal{F}(\mathbb{F}_p^n, \mathbb{F}_p))^k$ (each component written in $R_{p,n}$) be

$$\tilde{f}_t(x_1, \dots, x_n) = \sum_{i_1, \dots, i_n=0}^{p-1} a_{i_1, \dots, i_n}^{(t)} \prod_{j=1}^n x_j^{i_j},$$

for $1 \leq t \leq k$. Then

$$\deg_a \underline{f} = \max\{i_1 + \dots + i_n : a_{i_1, \dots, i_n}^{(t)} \neq 0, 1 \leq t \leq k\}$$

3.5. An explanation of change of basis and linear transformations in GL.

Our main aim is the analysis of the functions

$$f : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$$

for prime p and integers k, n with $k \leq n$. We naturally choose

$$e_1 = (1, 0, 0, \dots, 0), e_2 = (0, 1, 0, \dots, 0), \dots, e_n = (0, 0, \dots, 0, 1)$$

as a basis so that n -tuples and *coordinate vectors* are the same. However, we may want to choose another basis, say

$$b_1 = (1, 1, 1, \dots, 1), b_2 = (0, 1, 1, \dots, 1), \dots, b_n = (0, 0, \dots, 0, 1)$$

and write our n -tuples in this new basis. For instance the n -tuple $(1, 1, 1, 0, \dots, 0)$ corresponds to $b_1 - b_4$ and to the coordinate vector $(1, 0, 0, -1, 0, \dots, 0)$. Any non-singular $n \times n$ $\mathbb{F}_p$ -matrix (i.e., in $\text{GL}(n, p)$) corresponds to a basis change of $\mathbb{F}_p^n$. Thus the function

$$g := L_1 \circ f \circ L_2, \quad L_1 \in \text{GL}(k, p), L_2 \in \text{GL}(n, p)$$

is *equivalent* to the function f (we write $f \sim g$) as they represent the same function where different bases are chosen for the domain and codomain.

It is obvious that the operations $L_1 \circ f$ and $f \circ L_2$ preserve the algebraic degree of f .

3.6. How to move between $\mathcal{F}(\mathbb{F}_{p^m}^n, \mathbb{F}_{p^m})$ and $(\mathcal{F}(\mathbb{F}_p^{mn}, \mathbb{F}_p))^m$.

We will consider the case $n = 1$. The general case is straightforward.

Let

$$\tilde{f} \in R_{p^m, 1} \cong \mathcal{F}(\mathbb{F}_{p^m}, \mathbb{F}_{p^m})$$

be

$$\tilde{f}(x) = \sum_{j=0}^{p^m-1} c^{(j)} x^j.$$

Choose two bases $A = \{\alpha_1, \dots, \alpha_m\}$ and $B = \{\beta_1, \dots, \beta_m\}$. Set

$$x = \sum_{i=1}^m \alpha_i x_i, \text{ and } c^{(j)} = \sum_{i=1}^m \alpha_i c_i^{(j)},$$

where x_i and c_i are F_p variables and coefficients respectively, and plug this into the formula for $\tilde{f}(x)$, i.e.,

$$f(\underline{x}) = \sum_{j=0}^{p^m-1} \left(\sum_{i=1}^m \alpha_i c_i^{(j)} \right) \left(\sum_{i=1}^m \alpha_i x_i \right)^j.$$

Write every j in base- p , i.e., $j = j_{m-1}p^{m-1} + \dots + j_0$ with $0 \leq j_k < p$ and get

$$(1) \quad f(\underline{x}) = \sum_{j=0}^{p^m-1} \left(\sum_{i=1}^m \alpha_i c_i^{(j)} \right) \prod_{k=0}^{m-1} \left(\sum_{i=1}^m \alpha_i^{p^k} x_i \right)^{j_k}.$$

Rewriting all $\mathbb{F}_{p^m}$ coefficients that appear in the formula in B -basis, we get

$$f(\underline{x}) = f_1(\underline{x})\beta_1 + \dots + f_m(\underline{x})\beta_m,$$

where $f_i \in R_{p,m} \cong \mathcal{F}(\mathbb{F}_p^m, \mathbb{F}_p)$. The expansions

$$\left(\sum_{i=1}^m \alpha_i^{p^k} x_i \right)^{j_k}$$

are done using binomial expansion (in fact, multinomial), which is

$$(x + y)^{j_k} = \sum_{i=0}^{j_k} \binom{j_k}{i} x^i y^{j_k-i}.$$

Notice that $\binom{j_k}{i} \not\equiv 0 \pmod{p}$. Actually that is one of the reasons why we want to express everything in $\mathbb{F}_p$ (if $d \geq p$ then $p \mid \binom{d}{i}$ for certain i).

We write

$$\overline{f} = (f_1, \dots, f_m),$$

and say that $\overline{f}$ represents $\tilde{f}$ in the chosen A (for domain), and B (for codomain) bases. Similar to the previous section (since fields are vector spaces) base changes correspond to nonsingular linear transformations. Thus, $\overline{f'}$ which represents $\tilde{f}$ in bases A' and B' satisfies

$$\overline{f'} = L_1 \circ \overline{f} \circ L_2,$$

for some $L_1, L_2 \in \text{GL}(m, p)$.

Remark 3.7. To see the connection between a basis change and a non-singular linear transformation explicitly, consider the associativity of matrix multiplication

$$(u_1, \dots, u_m) \cdot (A_{m \times m} \cdot (\beta_1, \dots, \beta_m)) = ((u_1, \dots, u_m) \cdot A_{m \times m}) \cdot (\beta_1, \dots, \beta_m)$$

where the LHS corresponds to basis change and the RHS corresponds to a linear transformation of the coordinate vectors.

(Please read the Chapter 3 in `linalg.pdf` in the shared folder, if you want to refresh the linear algebra required to understand base changes.)

Let for $1 \leq t \leq m$,

$$f_t(x_1, \dots, x_m) = \sum_{i_1, \dots, i_m=0}^{p-1} a_{i_1, \dots, i_m}^{(t)} \prod_{j=1}^m x_j^{i_j}.$$

We usually combine all coefficients together and write

$$(2) \quad \bar{f}(x_1, \dots, x_m) = \sum_{i_1, \dots, i_m=0}^{p-1} a_{i_1, \dots, i_m} \prod_{j=1}^m x_j^{i_j},$$

where

$$a_{i_1, \dots, i_n} = (a_{i_1, \dots, i_n}^{(1)}, \dots, a_{i_1, \dots, i_n}^{(m)}).$$

3.7. Recovering algebraic degree from $R_{p^m,1}$ -representation. Since the bases we fix when we move from the $\mathbb{F}_q$ -representation to $\mathbb{F}_p$ -representation do not have an effect on the algebraic degree (see the previous subsection) we can talk of the algebraic degree of $\tilde{f} \in R_{p^m,1} \cong \mathcal{F}(\mathbb{F}_{p^m}, \mathbb{F}_{p^m})$ which can be defined by the algebraic degree of any $\mathbb{F}_p$ -representation $\bar{f} \in (R_{p,m})^m \cong (\mathcal{F}(\mathbb{F}_{p^m}, \mathbb{F}_p))^m$ of $\tilde{f}$. This value can be recovered directly from the $\mathbb{F}_q$ -representation.

Proposition 3.8. *Let $\tilde{f} \in \mathcal{F}(\mathbb{F}_{p^m}, \mathbb{F}_{p^m})$ be*

$$\tilde{f}(x) = \sum_{d=0}^{p^m-1} a_d x^d.$$

Then

$$\text{dega } \tilde{f} = \max\{\text{wt}_p(d) : a_d \neq 0\},$$

where the weight of the base- p representation $(d_{m-1}, \dots, d_0)_p$ of $0 \leq d \leq p^m - 1$ is the integer summation

$$\text{wt}_p(d) = \sum_{i=0}^{m-1} d_i.$$

Proof. From (1) and the binomial expansion formula we see that $\text{dega } \tilde{f} \leq \max\{\text{wt}_p(d) : a_d \neq 0\}$. Observe the one-to-one correspondence between the monomials $x^i, 0 \leq i \leq p^m - 1$ in $R_{p^m,1}$ and $x_1^{i_1} x_2^{i_2} \cdots x_m^{i_m}, 0 \leq i_1, i_2, \dots, i_m \leq p - 1$ in $(R_{p,m})^m$ that satisfy $\text{wt}_p(i) = i_1 + i_2 + \cdots + i_m$ naturally. Thus, inductively, using the upper bound on $\text{dega } \tilde{f}$, every degree-0 function $\underline{a} \in (R_{p,m})^m$ correspond to $\alpha x^0 \in R_{p^m,1}$, every degree-1 function $\sum_{i=1}^m \underline{a}_i x_i + \underline{a} \in (R_{p,m})^m$ correspond to $\sum_{i=0}^{m-1} \alpha_i x^{p^i} + \alpha$. Assume this holds for all degrees $< e$ and for degree- e we have $\text{dega } \tilde{f} < \max\{\text{wt}_p(d) : a_d \neq 0\}$. But this is a contradiction since all the degree $d - 1$ functions are covered by the induction hypothesis and by Lagrange interpolation two representations are unique (or, equivalently, $f_1 - f_2$ would be a degree- e 0-function). $\square$

4. BENT AND PERFECT NONLINEAR (PN) FUNCTIONS

4.1. Cryptographic functions and the notions of nonlinearity. In symmetric cryptography S-Boxes are of extreme importance. An S-Box is viewed as a **(vectorial) Boolean function**

$$F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m.$$

We also use the terminology **(n, m) -function**. We are almost always interested in the case $n \geq m$ and mostly in the extreme cases $m = 1$ and $m = n$. When $m = 1$, we have the convention to use small letters

$$f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2,$$

and say that f is a **Boolean function** or an **$(n, 1)$ -function**, and reserve capital letters for the “vectorial” case $m > 1$. Note that linear combinations of vectorial functions

$$F_u(x) := \langle u, F(x) \rangle, \quad u \in \mathbb{F}_2^m$$

are Boolean functions.

Recall that the **dot product** is a non-degenerate symmetric bilinear form $\mathbb{F}_2^n \times \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ defined as

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i,$$

where $x, y \in \mathbb{F}_2^n$ and $x = (x_1, x_2, \dots, x_n)$. Instead of the dot product, one can use any non-degenerate bilinear form to generate all linear combinations F_u of a vectorial Boolean function F . See [1, Chapter 5] for more details on bilinear forms.

4.2. Linear functions. A linear Boolean function $l : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ (also called a linear form or a linear functional) is a Boolean function that satisfy

$$l(x) + l(y) = l(x + y)$$

for all $x, y \in \mathbb{F}_2^n$.

In general, let V be an n -dimensional $\mathbb{K}$ -vector space. Then a $\mathbb{K}$ -linear form of V are functions $l : V \rightarrow \mathbb{K}$ that satisfy

$$\begin{aligned} l(x) + l(y) &= l(x + y), \quad x, y \in V, \\ l(cx) &= cl(x), \quad c \in \mathbb{K}, x \in V. \end{aligned}$$

Note that in the Boolean case we do not need to check the latter property of linear forms $l(cx) = cl(x)$ for all $c \in \mathbb{F}_2, x \in \mathbb{F}_2^n$, which holds trivially in the Boolean case. See [1, Chapter 5].

Every linear Boolean function can be written as

$$l_v(x) = \langle v, x \rangle, \quad v \in \mathbb{F}_2^n.$$

An **affine** Boolean function is of the form $x \mapsto l_v(x) + \epsilon$ where $\epsilon \in \mathbb{F}_2$.

4.3. Linear cryptanalysis. Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ be a vectorial Boolean function and define for $(u, v) \in \mathbb{F}_2^m \times \mathbb{F}_2^n$ and $u \neq 0$,

$$\lambda_F(u, v) = \#\{x \in \mathbb{F}_2^n : \langle u, F(x) \rangle = \langle v, x \rangle\} - 2^{n-1}.$$

Note that λ_F measures how much information linear combinations of input and output bits leak and related with the Hamming distance of the component functions F_u to all linear functions l_v .

Recall the notion of **Hamming distance** of two vectors.

Definition 4.1. Let $u, v \in \mathbb{F}_2^n$. The Hamming distance between u and v is defined as

$$d_H(u, v) = \#\{1 \leq i \leq n : u_i \neq v_i\}.$$

The Hamming weight of u is $\text{wt}_H(u) = d_H(u, 0)$.

The notion is extended in a straightforward manner to two Boolean functions $f, g : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$:

$$d_H(f, g) = \#\{x \in \mathbb{F}_2^n : f(x) \neq g(x)\}.$$

If for a carefully chosen set of (u, v) 's, the values $|\lambda_F(u, v)|$ are all *high*, one can devise a cryptanalysis of a cipher where the S-Box F is used in several consecutive rounds, called **linear cryptanalysis**.

Definition 4.2 (Nonlinearity). The **nonlinearity** of a vectorial Boolean function is defined as

$$\text{NL}_F = 2^{n-1} - \max\{|\lambda_F(u, v)| : (u, v) \in \mathbb{F}_2^m \times \mathbb{F}_2^n, u \neq 0\}.$$

4.3.1. Walsh transform. Relevant to the concept is the **Walsh transform** of a Boolean function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, defined as

$$\widehat{f}(v) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x)} (-1)^{\langle v, x \rangle},$$

which is the **Fourier transform** of $\tilde{f} : \mathbb{F}_2^n \rightarrow \{-1, 1\}$, where $\tilde{f} = (-1)^f$. We naturally extend the definition to cover vectorial Boolean functions $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ as (for $u \in \mathbb{F}_2^m, v \in \mathbb{F}_2^n$),

$$\widehat{F}(u, v) = \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle u, F(x) \rangle + \langle v, x \rangle},$$

which can be seen as the collection of the Walsh transforms of the **component Boolean functions** $F_u = \langle u, F \rangle$.

Exercise 4.3. Show that $\widehat{F}(u, v) = 2\lambda_F(u, v)$.

Question 4.4. What is the optimal nonlinearity of an (n, m) -function?

We are going to investigate the cases:

- First $m = 1$, and then
- $m = n$, and also some other cases.

4.4. Differential cryptanalysis. Define for a vectorial Boolean function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$,

$$\delta_F(a, b) = \#\{x \in \mathbb{F}_2^n : F(x + a) + F(x) = b\}.$$

Definition 4.5. A vectorial Boolean function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is called **perfect nonlinear** if

$$\delta_F(a, b) = 2^{n-m},$$

for all $(a, b) \in \mathbb{F}_2^n \times \mathbb{F}_2^m$ with $a \neq 0$.

The **differential uniformity** of F is defined by

$$\Delta_F = \max\{\delta_F(a, b) : (a, b) \in \mathbb{F}_2^n \times \mathbb{F}_2^m, a \neq 0\}.$$

Thus, perfect nonlinear functions have the best differential uniformity. Beware that the nomenclature is quite confusing. By definition, perfect nonlinear functions do not automatically mean they have the best nonlinearity as defined previously.

Similarly to the linear cryptanalysis, if for a carefully chosen set of (a, b) 's the values $\delta_F(a, b)$ are all *high*, one can devise a cryptanalysis of a cipher where the S-Box F is used in several consecutive rounds, called **differential cryptanalysis**.

Let us identify the worst possible vectorial Boolean functions for which $\delta_F(a, b)$ are maximal. These are of course, $\mathbb{F}_2$ -linear vectorial Boolean functions $\mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$. Similar to $\mathbb{F}_2$ -linear $(n, 1)$ -functions, these (n, m) -functions satisfy

$$L(x) + L(y) = L(x + y)$$

for all $x, y \in \mathbb{F}_2^n$ and can be described as an easy generalization

$$L(x_1, x_2, \dots, x_n) = \sum_{i=1}^n a^{(i)} x_i, \quad a^{(i)} \in \mathbb{F}_2^m.$$

Observe that this is the matrix multiplication $\mathbf{A}x$ where $\mathbf{A}$ is an $m \times n$ matrix with $\mathbb{F}_2$ -coefficients whose columns are $a^{(i)}$ and x is viewed as a column $(n \times 1)$ vector. It is clear that $\delta_L(a, L(a)) = 2^n$. This shows that, actually, differential uniformity is a way to measure how much F is *different* from linear (n, m) -functions.

Similarly, if a component function of F is $\mathbb{F}_2$ -linear, then the non-linearity of the function is the worst possible. Of course, if F is an $\mathbb{F}_2$ -linear (n, m) -function then all of its component functions are linear forms (the linear combinations of rows of $\mathbf{A}$).

An affine (n, m) -function is a linear (n, m) -function plus a constant (from $\mathbb{F}_2^m$). One can easily see that affine functions are (equally) as weak, from these points of view, as linear functions.

We can ask the analogous question.

Question 4.6. What is the optimal differential uniformity of an (n, m) -function?

We are going to investigate the cases:

- First $m = 1$, and then
- $m = n$, and also some other cases.

4.5. Walsh transform of Boolean functions. We start with the definition of the Walsh transform.

Definition 4.7. Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$. Define for $u \in \mathbb{F}_2^k, v \in \mathbb{F}_2^n$,

$$\widehat{f}(u, v) = \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle u, f(x) \rangle + \langle v, x \rangle}.$$

Bent functions are optimal functions in terms of their Walsh transform, i.e., the maximal absolute value of their Walsh transform is minimal. This follows Proposition 4.9 below.

Definition 4.8. Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$. We say that f is bent if

$$\widehat{f}(u, v) = \pm 2^{n/2}, \quad \forall u \in \mathbb{F}_2^k \setminus \{0\}, \forall v \in \mathbb{F}_2^n.$$

Observe that if $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$ is bent, then $f_u(x) := \langle u, f(x) \rangle$ is a function $f_u : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ and it is bent if and only if $u \in \mathbb{F}_2^k \setminus \{0\}$. Observe also that n has to be even for f to be bent.

Recall the following **power-moments** for Boolean function. When $k = 1$, we simply write $\widehat{f}(v) = \widehat{f}(1, v)$.

Proposition 4.9. Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$. Define the integer sum

$$S_f^{(d)} := \sum_{v \in \mathbb{F}_2^n} (\widehat{f}(v))^d.$$

Then

- (i) if $d = 1$ then $S_f^{(d)} = (-1)^{f(0)} 2^n$, and
- (ii) if $d = 2$ then $S_f^{(d)} = 2^{2n}$.

Proof. Follows from the double summation when the definition of the Walsh transform employed in the equation and then applying the Annihilator Lemma below. $\square$

Lemma 4.10 (Annihilator Lemma). *We have for $u \in \mathbb{F}_2^n$,*

$$\sum_{x \in \mathbb{F}_2^n} (-1)^{\langle u, x \rangle} = \begin{cases} 2^n & \text{if } u = 0, \\ 0 & \text{if } u \neq 0. \end{cases}$$

The following is the simplest example of a bent function on $\mathbb{F}_2^n \rightarrow \mathbb{F}_2$ when $\mathbb{F}_2^n$ is viewed as $\mathbb{F}_2^m \times \mathbb{F}_2^m$ where $n = 2m$.

Example 4.11 (The quadratic bent function). Let $f_{\text{quad}} : \mathbb{F}_2^m \times \mathbb{F}_2^m \rightarrow \mathbb{F}_2$ be defined as

$$f_{\text{quad}}(x, y) = \langle x, y \rangle.$$

Then f is bent.

Recall that the Walsh transform is related to the linear cryptanalysis. We have also another notion for Boolean functions that is related with the differential cryptanalysis. Here we allow odd p as well.

Definition 4.12 (Derivative). Let $f : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$. The derivative of f at $u \in \mathbb{F}_p^n$ is the function

$$D_f^{(u)} : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k,$$

defined as

$$D_f^{(u)} : x \mapsto f(x + u) - f(x).$$

The following definition is the generalization of bijectivity for functions $f : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$.

Definition 4.13 (Balance). We say that a function $f : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$ is balanced if

$$\#\{x \in \mathbb{F}_p^n : f(x) = a\} = p^{n-k}, \quad \forall a \in \mathbb{F}_p^k.$$

Now we can define perfect nonlinear functions that are optimal against differential cryptanalysis.

Definition 4.14 (Perfect nonlinear). Let $f : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$. We say that f is perfect nonlinear (PN) if the derivatives $D_f^{(u)}$ are balanced for all $u \in \mathbb{F}_p^n \setminus \{0\}$.

Example 4.15 (The quadratic bent function). Let $f_{\text{quad}} : \mathbb{F}_2^m \times \mathbb{F}_2^m \rightarrow \mathbb{F}_2$ be defined as

$$f_{\text{quad}}(x, y) = \langle x, y \rangle.$$

Then f is perfect nonlinear.

4.6. Bent $\iff$ PN. In this part, we will prove a theorem showing that “perfect” functions with respect to either notion of nonlinearity coincide.

Let us start with a theorem characterizing balance of functions in terms of the Walsh transform.

Define for any function $f : S \rightarrow T$ and $t \in T$,

$$f^{-1}(t) := \{s \in S : f(s) = t\}.$$

Theorem 4.16. *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$. Then f is balanced if and only if $\widehat{f}(u, 0) = 0$ for all $u \in \mathbb{F}_2^k \setminus \{0\}$.*

Proof. If f is balanced then $|f^{-1}(y)| = 2^{n-k}$ for all $y \in \mathbb{F}_2^k$. Now, for all $u \in \mathbb{F}_2^k \setminus \{0\}$,

$$\widehat{f}(u, 0) = \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle u, f(x) \rangle} = 2^{n-k} \sum_{y \in \mathbb{F}_2^k} (-1)^{\langle u, y \rangle} = 0,$$

by the Annihilator lemma.

For the converse, define for all $v \in \mathbb{F}_2^k$,

$$N_f(v) := \sum_{x \in \mathbb{F}_2^n} \sum_{u \in \mathbb{F}_2^k} (-1)^{\langle u, f(x) - v \rangle}.$$

Observe that the inner sum satisfies

$$\sum_{u \in \mathbb{F}_2^k} (-1)^{\langle u, f(x) - v \rangle} = \begin{cases} 2^k & \text{if } f(x) = v, \\ 0 & \text{if } f(x) \neq v. \end{cases}$$

Thus

$$N_f(v) = 2^k \# \{x \in \mathbb{F}_2^n : f(x) = v\} = 2^k |f^{-1}(v)|,$$

hence, f is balanced $\iff N_f(v) = 2^n$ for all $v \in \mathbb{F}_2^k$. Now, changing the order of summation,

$$\begin{aligned} N_f(v) &= \sum_{u \in \mathbb{F}_2^k} \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle u, f(x) - v \rangle} \\ &= \sum_{u \in \mathbb{F}_2^k} \widehat{f}(u, 0) \\ &= 2^n, \end{aligned}$$

by our assumption that $\widehat{f}(u, 0) = 0$ for all nonzero u and the obvious fact that $\widehat{f}(0, 0) = 2^n$. $\square$

Recall that for a function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, we defined the **d -th power moment** of the Walsh transform in Proposition 4.9 as

$$S_f^{(d)} := \sum_{v \in \mathbb{F}_2^n} (\widehat{f}(v))^d.$$

We now prove a proposition on the 4-th power moment of a Boolean function and show that the “best” value corresponds to the case of bent function whereas the “worst” value corresponds to the case of affine functions.

Proposition 4.17. *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$. Then*

$$2^{3n} \leq S_f^{(4)} \leq 2^{4n},$$

with equality on the lower bound if and only if f is bent and with equality on the upper bound if and only if f is affine.

Proof. Define for all $u \in \mathbb{F}_2^n$, the integer

$$\delta_f(u) := (\widehat{f}(u)^2) - 2^n.$$

We have by the 2nd power moment of the Walsh transform for Boolean functions (see Proposition 4.9)

$$S_f^{(2)} = 2^{2n}.$$

Thus

$$\sum_{u \in \mathbb{F}_2^n} \delta_f(u) = 0.$$

We have

$$\begin{aligned} S_f^{(4)} &= \sum_{u \in \mathbb{F}_2^n} ((\widehat{f}(u))^2)^2, \\ &= \sum_{u \in \mathbb{F}_2^n} (2^n + \delta_f(u))^2, \\ &= \sum_{u \in \mathbb{F}_2^n} 2^{2n} + \sum_{u \in \mathbb{F}_2^n} \delta_f(u)^2 + 2^{n+1} \sum_{u \in \mathbb{F}_2^n} \delta_f(u), \\ &= \sum_{u \in \mathbb{F}_2^n} 2^{2n} + \sum_{u \in \mathbb{F}_2^n} \delta_f(u)^2. \end{aligned}$$

That is to say $S_f^{(4)} \geq 2^{2n}$ with equality if and only if $\delta_f(u) = 0$ for all u , i.e., $\widehat{f}(u)^2 = 2^n$ that is to say f is bent.

One can show that the largest value one can get is when $\widehat{f}(u)^2 = 0$ for all u except for exactly one u_0 for which $\widehat{f}(u_0)^2 = 2^{2n}$, i.e., $\widehat{f}(u_0) = \pm 2^n$. To see this, let $a_u = \widehat{f}(u)^2$. We know that (again, by Proposition 4.9)

$$\sum_{u \in \mathbb{F}_2^n} a_u = A,$$

where $A = 2^n$. Observe that

$$\sum_{u \in \mathbb{F}_2^n} a_u^2 \leq A^2,$$

that is to say, the sum of squares of non-negative integers a_u is less than or equal to the square A^2 of the sum A . Equality happens only when $a_{u_0} = A$ for some u_0 and $a_u = 0$ when $u \neq u_0$.

Now this happens if and only if f is affine, i.e., $f(x) = \langle u_0, x \rangle + \epsilon$ for some $u_0 \in \mathbb{F}_2^n$ and $\epsilon \in \mathbb{F}_2$, since $\widehat{f}(u_0) = \pm 2^n$ means $f(x)$ agrees (or disagrees) with $\langle u_0, x \rangle$ for every x . $\square$

We now prove a similar proposition for the derivatives.

Proposition 4.18. *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ and define*

$$\mathcal{T}_f^{(2)} := \sum_{u \in \mathbb{F}_2^n} \left(\sum_{x \in \mathbb{F}_2^n} (-1)^{D_f^{(u)}(x)} \right)^2.$$

Then,

$$2^{2n} \leq \mathcal{T}_f^{(2)} \leq 2^{3n},$$

with equality on the lower bound if and only if f is bent and with equality on the upper bound if and only if f is affine.

Proof. A function f is affine if and only if $D_f^{(u)}$ is constant for all $u \in \mathbb{F}_2^n$ by definition. Thus the upper bound follows. The lower bound follows from the definition of bentness which stipulates all derivatives are balanced that is to say the sum

$$\left(\sum_{x \in \mathbb{F}_2^n} (-1)^{D_f^{(u)}(x)} \right)^2 = 0,$$

is minimum for all u . $\square$

Theorem 4.19 (Bent $\iff$ PN). *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$. Then f is bent if and only if f is PN.*

Proof. If $k = 1$ then the result follows from the two propositions we have just proved since

$$\begin{aligned} S_f^{(4)} &= \sum_{u \in \mathbb{F}_2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + \langle u, x \rangle} \sum_{y \in \mathbb{F}_2^n} (-1)^{f(y) + \langle u, y \rangle} \sum_{z \in \mathbb{F}_2^n} (-1)^{f(z) + \langle u, z \rangle} \sum_{t \in \mathbb{F}_2^n} (-1)^{f(t) + \langle u, t \rangle} \\ &= \sum_{x, y, z, t \in \mathbb{F}_2^n} (-1)^{f(x) + f(y) + f(z) + f(t)} \sum_{u \in \mathbb{F}_2^n} (-1)^{\langle u, x + y + z + t \rangle} \\ &= 2^n \sum_{x, y, z \in \mathbb{F}_2^n} (-1)^{f(x) + f(y) + f(z) + f(x + y + z)} \\ &= 2^n \sum_{x, y, u \in \mathbb{F}_2^n} (-1)^{f(x) + f(x + u) + f(y) + f(y + u)} \\ &= 2^n \mathcal{D}_f^{(2)}, \end{aligned}$$

using Annihilator lemma (i.e., $t = x + y + z$ is the only nonzero inner sum), making the change of variable $z = z + u$, and the definitions of $S_f^{(4)}$ and $\mathcal{D}_f^{(2)}$.

Recall that for $u \in \mathbb{F}_2^k$, f_u denotes the map $f_u : x \mapsto \langle u, f(x) \rangle$. For the general k observe that

$$\begin{aligned}
f \text{ is PN} &\iff D_f^{(a)} \text{ is balanced } \forall a \in \mathbb{F}_2^n \setminus \{0\} \\
&\iff x \mapsto \langle u, D_f^{(a)}(x) \rangle \text{ is balanced } \forall a \in \mathbb{F}_2^n \setminus \{0\}, \forall u \in \mathbb{F}_2^k \setminus \{0\} \\
&\iff D_{f_u}^{(a)} \text{ is balanced } \forall a \in \mathbb{F}_2^n \setminus \{0\}, \forall u \in \mathbb{F}_2^k \setminus \{0\} \\
&\iff f_u \text{ is bent } \forall u \in \mathbb{F}_2^k \setminus \{0\} \\
&\iff f \text{ is bent.}
\end{aligned}$$

The second line is Theorem 4.16 (observe that f_u is balanced if and only if $\widehat{f}(u, 0) = 0$) and the third line is by the bilinearity of $\langle, \rangle$. The rest is by definition. $\square$

Now let us give an upper bound on k , the maximal dimension for which a bent function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$ can exist.

Theorem 4.20. *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$ be bent. Then $k \leq n/2$.*

Proof. Let

$$f^{-1}(b) = \{x \in \mathbb{F}_2^n : f(x) = b\}.$$

We have by the Annihilator Lemma,

$$\#f^{-1}(b) = \frac{1}{2^m} \sum_{x \in \mathbb{F}_2^n, v \in \mathbb{F}_2^m} (-1)^{\langle v, f(x) + b \rangle}.$$

Now

$$\begin{aligned}
\#f^{-1}(b) &= \frac{1}{2^m} \sum_{x \in \mathbb{F}_2^n, v \in \mathbb{F}_2^m} (-1)^{\langle v, f(x) + b \rangle} \\
&= \frac{1}{2^m} \sum_{v \in \mathbb{F}_2^m} (-1)^{\langle v, b \rangle} \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle v, f(x) \rangle} \\
&= 2^n + \frac{1}{2^m} \sum_{v \in \mathbb{F}_2^m \setminus \{0\}} (-1)^{\langle v, b \rangle} \sum_{x \in \mathbb{F}_2^n} (-1)^{\langle v, f(x) \rangle} \\
&= 2^n + \frac{2^{n/2}}{2^m} \sum_{v \in \mathbb{F}_2^m \setminus \{0\}} (-1)^{\langle v, b \rangle + \gamma_v}
\end{aligned}$$

where $\gamma_v = \widehat{f}(v, 0)/2^{n/2} \in \mathbb{F}_2$ since f is bent. Since

$$\sum_{v \in \mathbb{F}_2^m \setminus \{0\}} (-1)^{\langle v, b \rangle + \gamma_v}$$

is odd (because sum of odd number of odd integers is odd), and $\#f^{-1}(b)$ is an integer, $2^{n/2-m}$ must be an integer, i.e., $m \leq n/2$. $\square$

4.7. Bent functions $f : \mathbb{F}_2^{2m} \rightarrow \mathbb{F}_2^m$. We will now show that there are (n, m) -bent functions for every $m \leq n/2$. This is equivalent to saying that there are (n, m) -PN functions for every $m \leq n/2$. We can prove the either property. In fact we will do both.

4.7.1. A nondegenerate bilinear form for finite fields. Let us first start with an important definition. The usual non-degenerate symmetric bilinear form $\mathbb{F}_2^m \times \mathbb{F}_2^m \rightarrow \mathbb{F}_2$ we use for vector spaces is the dot product $(x, y) \mapsto \langle x, y \rangle$. A non-degenerate symmetric bilinear form for $B : \mathbb{K}^m \times \mathbb{K}^m \rightarrow \mathbb{K}$ is the collection of all $\mathbb{K}$ -linear functionals in the sense that $l_x : y \mapsto B(x, y)$ (or, because of symmetry, equivalently $l_y : x \mapsto B(x, y)$) runs through all $\mathbb{K}$ -linear functionals $\mathbb{K}^m \rightarrow \mathbb{K}$.

Recall that we have two isomorphic ways to describe endomorphisms of the vector space $\mathbb{F}_p^m$. First one is by $\mathbf{M}_m(\mathbb{F}_p)$, i.e., left multiplication of column vectors by $m \times m$ $\mathbb{F}_p$ -matrices; and the second one is by $\mathcal{L}_{p,m}$, evaluation of the linearized polynomials at field elements. Thus, the linear functionals $x \mapsto \langle u, x \rangle$ may be identified as the matrix-vector multiplication

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} \mapsto \begin{pmatrix} u_1 & u_2 & \cdots & u_m \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}.$$

We need to identify those linearized polynomials

$$L_{\underline{\alpha}}(X) = \sum_{i=0}^{m-1} \alpha_i X^{p^i} \in \mathbb{F}_{p^m}[X],$$

with $\mathbb{F}_p$ -valued evaluation maps, i.e., $L_{\underline{\alpha}}(X) \in \mathbb{F}_p$ for all $X \in \mathbb{F}_{p^m}$. Thus, we must have

$$L_{\underline{\alpha}}^p - L_{\underline{\alpha}} \equiv 0 \pmod{X^{p^m} - X},$$

which implies $\alpha_i = \alpha_{i-1}^p$ for all $1 \leq i \leq m$ by simple manipulations where $\alpha_m = \alpha_0$. Thus, for every fixed $\alpha \in \mathbb{F}_{p^m}$, the maps

$$T_{\alpha} : X \mapsto \sum_{i=0}^{m-1} (\alpha X)^{p^i}$$

are $\mathbb{F}_p$ -valued, $\mathbb{F}_p$ -linear maps that form an m -dimensional $\mathbb{F}_p$ -vector space isomorphic to the vector space of linear functionals $x \mapsto \langle u, x \rangle$ for $u \in \mathbb{F}_p^m$. We define the trace map

$$\begin{aligned} \text{Tr} : \mathbb{F}_{p^m} &\rightarrow \mathbb{F}_p \\ X &\mapsto X + X^p + \cdots + X^{p^{m-1}}. \end{aligned}$$

Now it is straightforward to show that

$$B : (X, Y) \mapsto \text{Tr}(XY)$$

is a non-degenerate symmetric bilinear form since every $T_\alpha = X \mapsto B(X, \alpha) = \text{Tr}(\alpha X)$ defined above.

Theorem 4.21. *The function $f : \mathbb{F}_{2^m} \times \mathbb{F}_{2^m} \rightarrow \mathbb{F}_{2^m}$ with $f : (x, y) \mapsto xy$ is bent.*

Proof. In the following $M_x : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^m$ is the $\mathbb{F}_2$ -linear multiplication map $M_x : y \mapsto xy$. We will first prove the claim using the dot product, willfully forgetting the field structure:

$$\begin{aligned} \widehat{F}(u, (v, w)) &= \sum_{x, y \in \mathbb{F}_2^m} (-1)^{\langle u, F(x) \rangle + \langle (v, w), (x, y) \rangle} \\ &= \sum_{x, y \in \mathbb{F}_2^m} (-1)^{\langle u, M_x(y) \rangle + \langle v, x \rangle + \langle w, y \rangle} \\ &= \sum_{x \in \mathbb{F}_2^m} (-1)^{\langle v, x \rangle} \sum_{y \in \mathbb{F}_2^m} (-1)^{\langle u, M_x(y) \rangle + \langle w, y \rangle} \\ &= \sum_{x \in \mathbb{F}_2^m} (-1)^{\langle v, x \rangle} \sum_{y \in \mathbb{F}_2^m} (-1)^{\langle M_x^T(u) + w, y \rangle} \\ &= 2^m \sum_{x \in \mathbb{F}_2^m \text{ s.t. } M_x^T(u) = w} (-1)^{\langle v, x \rangle} \end{aligned}$$

For every fixed $0 \neq u, w \in \mathbb{F}_{2^m}$ there exists unique $x \in \mathbb{F}_{2^m}$ such that $M_x^T(u) = w$. To see this, define $L_u(x) = M_x^T(u)$. Since $M_x(u) + M_y(u) = M_{x+y}(u)$ (i.e., $xu + yu = (x + y)u$), we have, $M_x^T(u) + M_y^T(u) = M_{x+y}^T(u)$ (property of transpose). Therefore L_u is linear (by the definition of L_u). Since $L_u(x) = 0$ if and only if either $x = 0$ or $u = 0 \iff M_x^T(u) = 0$ if and only if either $x = 0$ or $u = 0 \iff M_x(u) = 0$ if and only if either $x = 0$ or $u = 0 \iff xu = 0$ if and only if $x = 0$ or $u = 0$; we must have L_u is bijective for all $0 \neq u \in \mathbb{F}_{2^m}$.

The above proof is awkward since we do not use the power of the vector space-field identification. Define

$$\widehat{F}_{\text{Tr}}(\alpha, (\beta, \gamma)) = \sum_{x, y \in \mathbb{F}_{2^m}} (-1)^{\text{Tr}(\alpha F(x, y)) + \text{Tr}(\beta x) + \text{Tr}(\gamma y)}.$$

Recall that when we run through all $v \in \mathbb{F}_2^m$ and $\beta \in \mathbb{F}_{2^m}$, $x \mapsto \langle v, x \rangle$ and $x \mapsto \text{Tr}(\beta x)$ runs through the same functionals, i.e., we have a one-to-one correspondence between vectors v and the field elements β (mapping 0-vector to the 0 of the field). Thus there is a one to one correspondence between $\widehat{F}(u, (v, w))$ and $\widehat{F}_{\text{Tr}}(\alpha, (\beta, \gamma))$.

Now,

$$\begin{aligned}
 \widehat{F}_{\text{Tr}}(\alpha, (\beta, \gamma)) &= \sum_{x, y \in \mathbb{F}_{2^m}} (-1)^{\text{Tr}(\alpha F(x, y)) + \text{Tr}(\beta x) + \text{Tr}(\gamma y)} \\
 &= \sum_{x, y \in \mathbb{F}_{2^m}} (-1)^{\text{Tr}(\alpha xy) + \text{Tr}(\beta x) + \text{Tr}(\gamma y)} \\
 &= \sum_{x \in \mathbb{F}_{2^m}} (-1)^{\text{Tr}(\beta x)} \sum_{y \in \mathbb{F}_{2^m}} (-1)^{\text{Tr}(\alpha xy + \gamma y)} \\
 &= \sum_{x \in \mathbb{F}_{2^m}} (-1)^{\text{Tr}(\beta x)} \sum_{y \in \mathbb{F}_{2^m}} (-1)^{\text{Tr}(y(\alpha x + \gamma))}.
 \end{aligned}$$

Now by Annihilator Lemma,

$$\widehat{F}_{\text{Tr}}(\alpha, (\beta, \gamma)) = 2^m (-1)^{\text{Tr}(\beta \gamma / \alpha)},$$

recalling that $\alpha \neq 0$.

□

Note that if there is a bent function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$ then there exists a bent function $g : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^{k-1}$ (which may be defined as the projection of f to an $n - 1$ -dimensional vector space).

The proof of this is quite obvious. Let $\{e_1, e_2, \dots, e_n\}$ be a basis of $\mathbb{F}_2^n$ and $f = (f_{e_1}, f_{e_2}, \dots, f_{e_n})$ be a bent function. Then, for instance, $g = (f_{e_1}, f_{e_2}, \dots, f_{e_{n-1}})$ is bent since f is bent $\iff$ for all $u \in \mathbb{F}_2^n$ the functions $f_u : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, $f_u : x \mapsto \langle u, f(x) \rangle = \sum_{i=1}^n u_i f_{e_i}(x)$ is bent. Thus, g is bent.

Remark 4.22. The number of $k - 1$ dimensional vector-subspaces of $\mathbb{F}_2^k$ is $2^k - 1$. They are all of the type

$$V_u = \{x \in \mathbb{F}_2^k : \langle u, x \rangle = 0\},$$

for $u \in \mathbb{F}_2^k \setminus \{0\}$.

We could have as well proved that f is PN which we will do now.

Theorem 4.23. *The function $f : \mathbb{F}_{2^m}^2 \rightarrow \mathbb{F}_{2^m}$ with $f : (x, y) \mapsto xy$ is perfect nonlinear.*

Proof. We need to show that the (normalized) derivatives

$$\Delta_{u,v} : (x, y) \mapsto f(x, y) + f(x + u, y + v) + f(u, v)$$

are balanced for fixed $(u, v) \neq (0, 0)$. Note that this is equivalent to saying

$$D_{u,v} : (x, y) \mapsto f(x, y) + f(x + u, y + v)$$

are balanced, which is our original definition of perfect nonlinearity. Now,

$$\Delta_{u,v}(x, y) = xv + yu.$$

Since this map is $\mathbb{F}_2$ -linear (it is in fact $\mathbb{F}_{2^m}$ -linear), it is enough to check that the $\mathbb{F}_2$ -dimension of its kernel is m . If $u = 0$ then $(x, y) \in \{0\} \times \mathbb{F}_{2^m}$, if $v = 0$ then $(x, y) \in \mathbb{F}_{2^m} \times \{0\}$, and otherwise $\{(x, xv/u) : x \in \mathbb{F}_{2^m}\}$ describes the kernels of the maps. Note that these m -dimensional $\mathbb{F}_2$ -vector spaces are in fact 1-dimensional $\mathbb{F}_{2^m}$ -vector spaces. $\square$

Remark 4.24. Observe that $(x, y) \mapsto (v, u)(x, y)^T = xv + yu$ are precisely the (all) $\mathbb{F}_{2^m}$ -linear functionals from $\mathbb{F}_{2^m} \times \mathbb{F}_{2^m} \rightarrow \mathbb{F}_{2^m}$.

4.8. The notion of equivalence for functions. The following is the basic notion of equivalence of functions which, we will show, preserves many cryptographically interesting properties.

Definition 4.25 (Extended-Affine equivalence). Let $f, g : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$. We say that f is EA-equivalent to g and write $f \sim g$ if there exist

- (i) an affine bijection $A_1 : \mathbb{F}_p^k \rightarrow \mathbb{F}_p^k$,
- (ii) an affine bijection $A_2 : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$, and
- (iii) an affine map $A_3 : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^k$, such that

$$g(x) = A_1 \circ f \circ A_2(x) + A_3(x).$$

The following proposition is easy to show.

Proposition 4.26. *We have the following facts for the EA-equivalence.*

- (i) $\sim$ is an equivalence relation.
- (ii) $\sim$ preserves the DDT.
- (iii) $\sim$ preserves the absolute values of the LAT.
- (iv) $\sim$ preserves the algebraic degree dega .
- (v) $\sim$ does not necessarily preserve balance.

The following theorem shows that we can classify quadratic functions $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ up to the EA-equivalence $\sim$.

Theorem 4.27. *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be quadratic ($\deg_a f = 2$). Then,*

$$f(x_1, x_2, \dots, x_n) \sim \sum_{i=1}^r x_{2i-1}x_{2i}$$

for some $1 \leq r \leq \lfloor n/2 \rfloor$ which is called the rank of f .

Proof of the theorem is clear by successive application of the following lemma.

Lemma 4.28. *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be quadratic. Then,*

$$f(x_1, x_2, \dots, x_n) \sim x_1x_2 + g(x_3, \dots, x_n),$$

where $\deg_a g \leq 2$.

Proof. We can, by adding affine functions and renaming variables if necessary, easily get

$$f \sim x_1x_2 + x_1a(x_3, x_4, \dots, x_n) + x_2b(x_3, x_4, \dots, x_n) + h(x_3, x_4, \dots, x_n),$$

where

$$a(x_3, x_4, \dots, x_n) = a_3x_3 + \dots + a_nx_n,$$

$$b(x_3, x_4, \dots, x_n) = b_3x_3 + \dots + b_nx_n,$$

$$h(x_3, x_4, \dots, x_n) = \sum_{3 \leq i < j \leq n} h_{ij}x_ix_j.$$

Apply the non-singular transformation

$$y_1 = x_1 + b(x_3, x_4, \dots, x_n),$$

$$y_2 = x_2 + a(x_3, x_4, \dots, x_n),$$

$$y_i = x_i \text{ for } i > 2.$$

Now

$$\begin{aligned} f &\sim (y_1 + b(y))(y_2 + a(y)) + (y_1 + b(y))a(y) + (y_2 + a(y))b(y) + h(y), \\ &= y_1y_2 + h(y) + b(y)a(y), \end{aligned}$$

which is our claim since h, a, b depend only on $y_3, \dots, y_n$. □

Corollary 4.29. *Let $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be quadratic ($\deg_a f = 2$). Then $\widehat{f}(v) \in \{0, \pm 2^{n-r}\}$.*

Prove the corollary as exercise.

5. APN FUNCTIONS

In the last section, we proved that $(n, n/2 + 1)$ -bent and $(n, n/2 + 1)$ -perfect nonlinear functions do not exist. In this section we will deal with the cryptographically interesting case of optimal (n, n) -functions. When we say optimal, we mean (n, n) -functions F that have *optimal* Δ_F and Λ_F values.

Let us start with the differential uniformity and Δ_F .

5.1. (n, n) -functions with optimal differential uniformity. Let F be an (n, n) -function. If the equality

$$F(x) + F(x + a) = b$$

has a solution $x = x_0$ then $x = x_0 + a$ is another solution. Therefore the smallest value we can get for Δ_F is

$$\Delta_F = \max\{\delta_F(a, b) : (a, b) \in \mathbb{F}_2^n \times \mathbb{F}_2^n, a \neq 0\} \geq 2.$$

Definition 5.1. An (n, n) -function F is called **almost perfect nonlinear (APN)** if $\Delta_F = 2$.

Define the derivative set

$$\text{Der}_F^{(a)} = \{F(x) + F(x + a) : x \in \mathbb{F}_2^n\}.$$

The following proposition provides an alternative definition of APN functions.

Proposition 5.2. *The (n, n) -function F is APN if and only if $|\text{Der}_F^{(a)}| = 2^{n-1}$ for all $a \neq 0$.*

Proof. This is easy to see from the definition. □

As was the case for $(n, n/2)$ -functions, finite field representation of vectorial Boolean functions allows a neat description of the known APN functions.

5.2. Power functions (monomials). Since we use the finite field representation for functions, polynomials from $\mathbb{F}_{2^n}[X]$ become the main focus. Thus, polynomials that have the simplest representation $F(X) = X^d$ become the first class of interest. The bijective behaviour of such functions is easy to determine.

Theorem 5.3. *Let $F(X) = aX^d + b$ where $a, b \in \mathbb{F}_{p^n}$ and $a \neq 0$. Then F permutes $\mathbb{F}_{p^n}$ if and only if $\gcd(d, p^n - 1) = 1$.*

Proof. We have $\mathbb{F}_{p^n}^\times = \langle \omega \rangle$ and $\omega^i \mapsto \omega^{id}$ permutes $\mathbb{F}_{p^n}^\times$ if and only if $\gcd(d, p^n - 1) = 1$. □

For monomials checking APNness is easier.

Proposition 5.4. *Let $F(x) = x^d$ be an (n, n) -function. Then F is APN if and only if $|\text{Der}_F^{(1)}| = 2^{n-1}$.*

Proof. For nonzero a , we have

$$\begin{aligned} \#\{x^d + (x+a)^d : x \in \mathbb{F}_{2^n}\} &= \#\{ay^d + (ay+a)^d : y \in \mathbb{F}_{2^n}\} \\ &= \#\{a^d(x^d + (x+1)^d) : x \in \mathbb{F}_{2^n}\}. \end{aligned}$$

□

Theorem 5.5. *Let $F(X) = X^3 \in \mathbb{F}_{2^n}[X]$. Then F is an APN function.*

Proof. It is enough to show $X \mapsto X^3 + (X+1)^3 = X^2 + X + 1$ is two-to-one. Note that $X^2 + X$ is a linearized polynomial. Thus $X \mapsto X^2 + X$ is an $\mathbb{F}_2$ -linear map. The kernel of this map contains 0 and 1. Since the polynomial degree of $X^2 + X$ is two, there are no more solutions to $X^2 + X = 0$. Thus $X \mapsto X^2 + X + 1$ is two-to-one. □

Exercise 5.6. Show that the (n, n) -function defined by $F(X) = X^{2^k+1}$ is APN if and only if $\gcd(k, n) = 1$.

Hint: You need to show

$$\gcd(2^k - 1, 2^n - 1) = 2^{\gcd(k, n)} - 1.$$

The following function is the cryptographically interesting inverse function which is used as an S-Box in the AES cipher.

Theorem 5.7. *Let $F(X) = X^{2^n-2}$ be an (n, n) -function. Then F is an APN function if and only if n is odd.*

Proof. Observe that for $X \in \mathbb{F}_{2^n}^\times$, the map $X \mapsto X^{2^n-2}$ is identical to $X \mapsto 1/X$. We have

$$\text{Der}_F^{(1)} = \{1\} \cup \{1/X + 1/(X+1) : X \in \mathbb{F}_{2^n} \setminus \{0, 1\}\}.$$

Note that $1/X + 1/(X+1) = 1/(X^2 + X)$. Thus, F is APN if and only if $X^2 + X = 1$ is not satisfied for some $X \in \mathbb{F}_{2^n} \setminus \{0, 1\}$ since $X \mapsto 1/(X^2 + X)$ is two-to-one on $\mathbb{F}_{2^n} \setminus \{0, 1\}$. Observe that the polynomial $X^2 + X + 1$ is irreducible over $\mathbb{F}_2[X]$ and both of its roots are in $\mathbb{F}_4 \setminus \mathbb{F}_2$. Since $\mathbb{F}_4 \subseteq \mathbb{F}_{2^n}$ if and only if n is even, we are done. □

5.3. Equivalences of (n, n) -functions. We can extend the extended-affine equivalence of $(n, 1)$ -functions to (n, n) -functions.

Definition 5.8. Let F, G be two (n, n) -functions. We say that F and G are EA-equivalent if there exists $\mathbb{F}_2$ -affine bijections $A_1, A_2 : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ and an $\mathbb{F}_2$ -affine

function $A_3 : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ such that

$$G(x) = A_1 \circ F \circ A_2(x) + A_3(x)$$

for an indeterminate x . In that case we write $F \sim_{EA} G$.

Exercise 5.9. Show that the EA -equivalence is indeed an equivalence relation and it preserves

- (i) the algebraic degree,
- (ii) the differential spectrum, and
- (iii) the absolute values of the Walsh spectrum.

Remark 5.10 (Compositional inverse of a permutation). Let F be a bijective (n, n) -function. We saw examples of bijective APN functions above. Let us explore the APNness of the compositional inverse F^{-1} of F .

Note that, the condition for APNness is equivalent to stipulating that the equations

$$x + y = a,$$

$$F(x) + F(y) = b,$$

have 0 or 2 common solutions for $0 \neq a \in \mathbb{F}_2^n$ and $b \in \mathbb{F}_2^n$. Let F be a bijective APN function. Introduce new variables z, t such that $F(x) = z$ and $F(y) = t$. Thus,

$$z + t = b,$$

$$F^{-1}(z) + F^{-1}(t) = a,$$

have 0 or 2 common solutions for $0 \neq a \in \mathbb{F}_2^n$ and $b \in \mathbb{F}_2^n$. The only thing we need to show is the case $a = 0$ and $b \neq 0$, which has no solutions since F^{-1} is bijective. Thus F^{-1} is APN.

Therefore, we deduce F is APN if and only if F^{-1} is APN.

A natural question is whether our notion of equivalence (EA) covers this natural construction, that is to say whether $F \sim_{EA} F^{-1}$. One can give a simple counterexample. Let $F(X) = X^3$ be a function on $\mathbb{F}_{2^5}$. It is readily seen that F is bijective since $\gcd(3, 31) = 1$. Then $F^{-1}(X) = X^{21}$, since $F \circ F^{-1}(X) = (X^{21})^3 = X^{63} \equiv X \pmod{X^{32} + X}$. Now, $\deg F = \text{wt}_2(3) = 2$ but $\deg F^{-1} = \text{wt}_2(21) = 3$. Thus $F \not\sim_{EA} F^{-1}$.

The following notion of equivalence generalizes EA -equivalence and preserves inversion. We define for an (n, n) -function F , the **graph** of F as

$$\Gamma_F = \{(x, F(x)) : x \in \mathbb{F}_2^n\}.$$

Definition 5.11. We say that two (n, n) -functions F, G are *CCZ-equivalent* if there exist four $\mathbb{F}_2$ -linear maps $A, B, C, D : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, such that

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

is full-rank, a bijection $\pi : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, and two constants $u, v \in \mathbb{F}_2^n$ such that

$$(3) \quad \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ F(x) \end{pmatrix} + \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \pi(x) \\ G(\pi(x)) \end{pmatrix},$$

for an indeterminate x .

Note that this means

$$\begin{aligned} \pi(x) &= A(x) + B \circ F(x) + u, \\ G(\pi(x)) &= C(x) + D \circ F(x) + v =: \rho(x), \end{aligned}$$

and, thus,

$$G = \rho \circ \pi^{-1}.$$

Proposition 5.12. *CCZ-equivalence preserves differential spectrum.*

Exercise 5.13. Show that *CCZ-equivalence* is indeed an equivalence relation and that

- (i) *CCZ-equivalence* preserves absolute values of the Walsh spectrum,
- (ii) if F is bijective then F^{-1} is *CCZ-equivalent* to F ,
- (iii) if $F \sim_{EA} G$ then $F \sim_{CCZ} G$.

5.4. APN functions and the Walsh transform. In the $(n, 1)$ -function case, functions with optimal differential and Walsh spectrum coincide as we have seen in Theorem 4.19. This is not true in the (n, n) -function case. Let us first give a characterization of APN functions by Walsh power sums.

Theorem 5.14. *An (n, n) -function F satisfy*

$$\sum_{0 \neq u, v \in \mathbb{F}_2^n} (\widehat{F}(u, v))^4 \geq 3 \cdot 2^{4n} - 2 \cdot 2^{3n}.$$

Moreover, F is APN if and only if equality holds.

Proof. After expansion, we simply get

$$\begin{aligned} \sum_{u, v \in \mathbb{F}_2^n} (\widehat{F}(u, v))^4 &= \sum_{x, y, z, t \in \mathbb{F}_2^n} \sum_{u \in \mathbb{F}_2^n} (-1)^{\langle u, F(x) + F(y) + F(z) + F(t) \rangle} \sum_{v \in \mathbb{F}_2^n} (-1)^{\langle v, x + y + z + t \rangle} \\ &= 2^n \sum_{x, y, z \in \mathbb{F}_2^n} \sum_{u \in \mathbb{F}_2^n} (-1)^{\langle u, F(x) + F(y) + F(z) + F(x + y + z) \rangle} \end{aligned}$$

by Annihilator Lemma. Noting that, writing $a = x + z$ we get,

$$\{F(x)+F(y)+F(z)+F(x+y+z) : x, y, z \in \mathbb{F}_2^n\} = \{F(x)+F(x+a)+F(y)+F(y+a) : x, y, a \in \mathbb{F}_2^n\},$$

and observing that F is APN if and only if $F(x) + F(x+a) = F(y) + F(y+a)$ holds only trivially, i.e., $a = 0, x = y$ or $x = y + a$, we prove the result. $\square$

An $(n, 1)$ -function f has optimal nonlinearity when

$$\max\{|\widehat{f}(v)| : v \in \mathbb{F}_2^n\} = 2^{n/2}.$$

An analogous result for (n, n) -functions is given below.

Theorem 5.15 (Sidelnikov-Chabaud-Vaudenay bound). *Let F be an (n, n) -function. Then*

$$\max\{|\widehat{F}(u, v)| : 0 \neq u, v \in \mathbb{F}_2^n\} \geq 2^{(n+1)/2}.$$

Proof. Observe that

$$Q_F \sum_{0 \neq u, v \in \mathbb{F}_2^n} (\widehat{F}(u, v)^2) \geq \sum_{0 \neq u, v \in \mathbb{F}_2^n} (\widehat{F}(u, v)^4)$$

where

$$Q_F := \max\{\widehat{F}(u, v)^2 : 0 \neq u, v \in \mathbb{F}_2^n\}.$$

The rest of the proof using Theorem 5.14 is left as an exercise. $\square$

When n is odd, the bound is achievable.

Definition 5.16. An (n, n) -function F is called **almost bent (AB)** if

$$\max\{|\widehat{F}(u, v)| : 0 \neq u, v \in \mathbb{F}_2^n\} = 2^{(n+1)/2}.$$

Exercise 5.17. Show that if F is AB then necessarily $\widehat{F}(u, v) \in \{0, \pm 2^{(n+1)/2}\}$.

Exercise 5.18. Show that F is AB implies F is APN. Note that necessarily n is odd.

Remark 5.19. There are APN functions that are not AB even when n is odd. An example is the inverse function.

REFERENCES

1. Peter J. Cameron, *Notes on Linear Algebra*, Lecture Notes, 2008.
